Feature engineering and supervised modeling

AI−use statement

I prepared preliminary code and a report draft, and then I used OpenAI Codex to help me revise and complete the R analysis, debug the code, and help interpret the results. Codex also helped expand and format the final report with figures and tables. AI saved time, but it could still make coding or statistical mistakes. I am responsible for reviewing and understanding the final work.

2.1 Modeling table and prediction target

I used set−a only: 4,000 intensive care unit (ICU) stays, including 554 in−hospital deaths (13.85%). Each row is one stay, and the script checks that RecordID is unique. The outcome is in−hospital death, joined by RecordID. Predictors summarize the first 48 ICU hours. This gives a prediction after the full observation window, not at admission.

Cleaning carried over from Checkpoint 1

The script rebuilds the long and wide tables from the original patient files on every run. It carries over the Checkpoint 1 cleaning rules. Missing codes of −1 are set to missing where appropriate. Original values remain in ValueRaw. Screening ranges include height 100−250 cm, weight 20−300 kg, pH 6.5−8.0, and temperature 20−45 degrees C. These rules also remove zero height and weight values. Some extreme but non−negative laboratory and vital−sign values remain. This is a limitation rather than proof they are correct.

What is excluded

RecordID is used only to identify stays and check splits. Hospital length of stay and survival time are excluded because they contain later outcome information. The Simplified Acute Physiology Score I (SAPS−I) and Sequential Organ Failure Assessment (SOFA) score are benchmarks, not inputs to the three models. No set−b or set−c files are read.

Data source: PhysioNet/Computing in Cardiology Challenge 2012, version 1.0.0.

2.2 Predictors

Table 1 groups the 194 input columns before training−fold zero−variance removal. ICU type is represented by indicator columns rather than treated as a continuous number. Gender also has an unknown category. The full column list is in cp2_predictor_catalog.csv.

Table 1. Predictor groups and clinical reasons

Group	Columns	Reason
Admission variables	9	Age, body size, sex, and ICU case mix
Last values	37	Condition nearest the end of the window
Minimum and maximum	74	Most abnormal states during the window
Mean values	37	Typical measured state across 48 hours
Measurement counts	37	Monitoring intensity and never measured status

Measurements included

Albumin, ALP, ALT, AST, Bilirubin, BUN, Cholesterol, Creatinine, DiasABP, FiO2, GCS, Glucose, HCO3, HCT, HR, K, Lactate, MAP, MechVent, Mg, Na, NIDiasABP, NIMAP, NISysABP, PaCO2, PaO2, pH, Platelets, RespRate, SaO2, SysABP, Temp, TroponinI, TroponinT, Urine, WBC, Weight . Each has a last value, minimum, maximum, mean, and count.

Missingness and preprocessing

No clinical variable is dropped for low coverage. Counts are transformed with log(1 + count), so zero means no valid measurement and high counts have less influence. First values are omitted to reduce redundancy. Missing numeric values are filled with the training−fold median. I chose medians because they are simple and less affected by extreme values. Columns with zero variance are removed using the training fold. Remaining columns are centered and scaled using that fold. The held−out fold never sets these values.

The generalized additive model (GAM) uses 12 clinically chosen continuous predictors, sex, ICU type, and lactate and ventilation measurement counts. This keeps the nonlinear model manageable.

2.3 Cross−validation and tuning

I used five−fold stratified outer cross−validation (CV), with the same stay−level folds for every method. Each patient receives one out−of−fold prediction per method. In each outer training set, three−fold stratified inner CV chooses hyperparameters by mean Brier score (mean squared error of predicted probability). Preprocessing is fitted again inside every inner training fold. This prevents imputation and scaling from leaking across either level.

Table 2. Prespecified tuning search

Model	Search
Elastic net	alpha = 0.5; 20 lambdas: 0.0005 to 0.2
GAM	12 cubic smooths, k = 4; gamma = 1 or 1.4
Random forest	300 trees; mtry = 14 or 48; min node = 10 or 30

Elastic net combines shrinkage with variable selection and is useful for correlated summaries of the same measurement. The generalized additive model (GAM) uses shrinkage smooths fitted by restricted maximum likelihood, with a fixed ridge penalty on linear terms. The forest uses median imputation, no early stopping, no oversampling, and no class weighting.

Table 3. Outer−fold audit

Fold	Training stays	Held−out stays	Held−out deaths	Shared IDs
1	3199	801	111	0
2	3200	800	111	0
3	3200	800	111	0
4	3200	800	111	0
5	3201	799	110	0

All folds keep stays separate. Random seed: 7072026. Inner folds are shared across the three models.

2.7 Comparing models

The area under the receiver operating characteristic (ROC) curve, or AUROC, measures ranking. A value of 0.5 is random ranking and 1 is perfect. Table 4 gives each fold, its mean, and standard deviation (SD). Missing SAPS−I or SOFA scores are filled with the outer training−set median for comparison on the same patients. These benchmarks are raw scores, not calibrated probabilities.

Table 4. Held−out AUROC by fold

Method	F1	F2	F3	F4	F5	Mean	SD
Elastic net	0.846	0.836	0.842	0.850	0.840	0.843	0.005
GAM	0.784	0.763	0.798	0.807	0.804	0.791	0.018
Random forest	0.852	0.815	0.838	0.869	0.819	0.839	0.023
SAPS−I	0.652	0.652	0.668	0.651	0.607	0.646	0.023
SOFA	0.655	0.599	0.640	0.618	0.586	0.620	0.028

Figure 1. ROC curves from held−out fold predictions

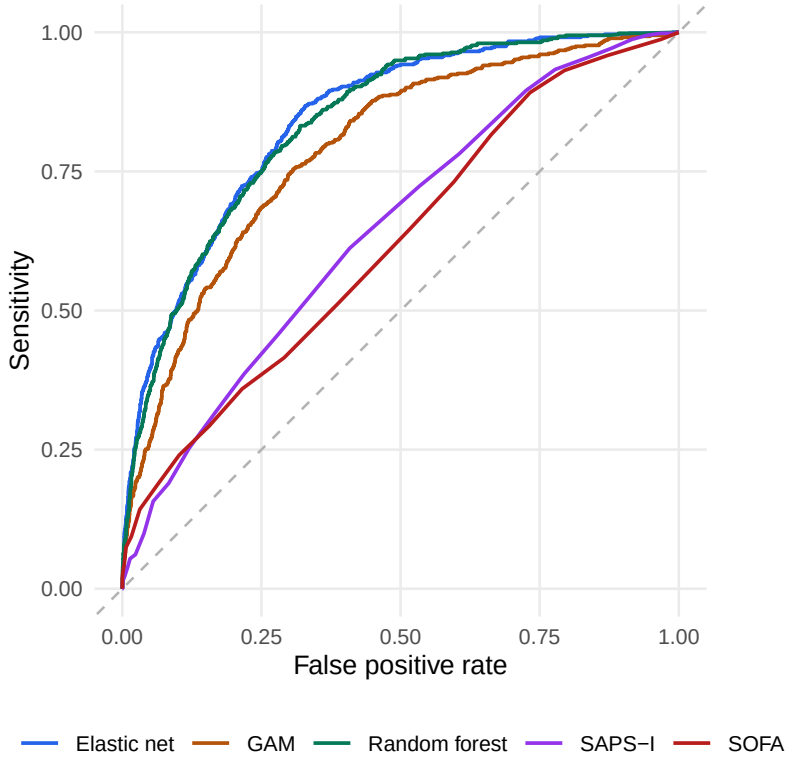


Figure 1 uses out−of−fold predictions for all five methods. Elastic net has the highest model mean AUROC (0.843). The best benchmark is SAPS−I (0.646); the gap is 0.197, compared with fold SDs of 0.005 and 0.023. The gain exceeds this fold variation, but this is not a significance test.

2.4 Regularized logistic regression

The refit elastic−net model uses alpha = 0.5 and lambda = 0.01171, selected by inner CV on all set−a. It retains 55 nonzero coefficients, listed in Appendix Table A1. Figure 2 shows the 18 largest on a common scale. Positive values increase predicted log odds, conditional on the other inputs; they do not identify treatment effects.

Figure 2. Largest elastic−net coefficients

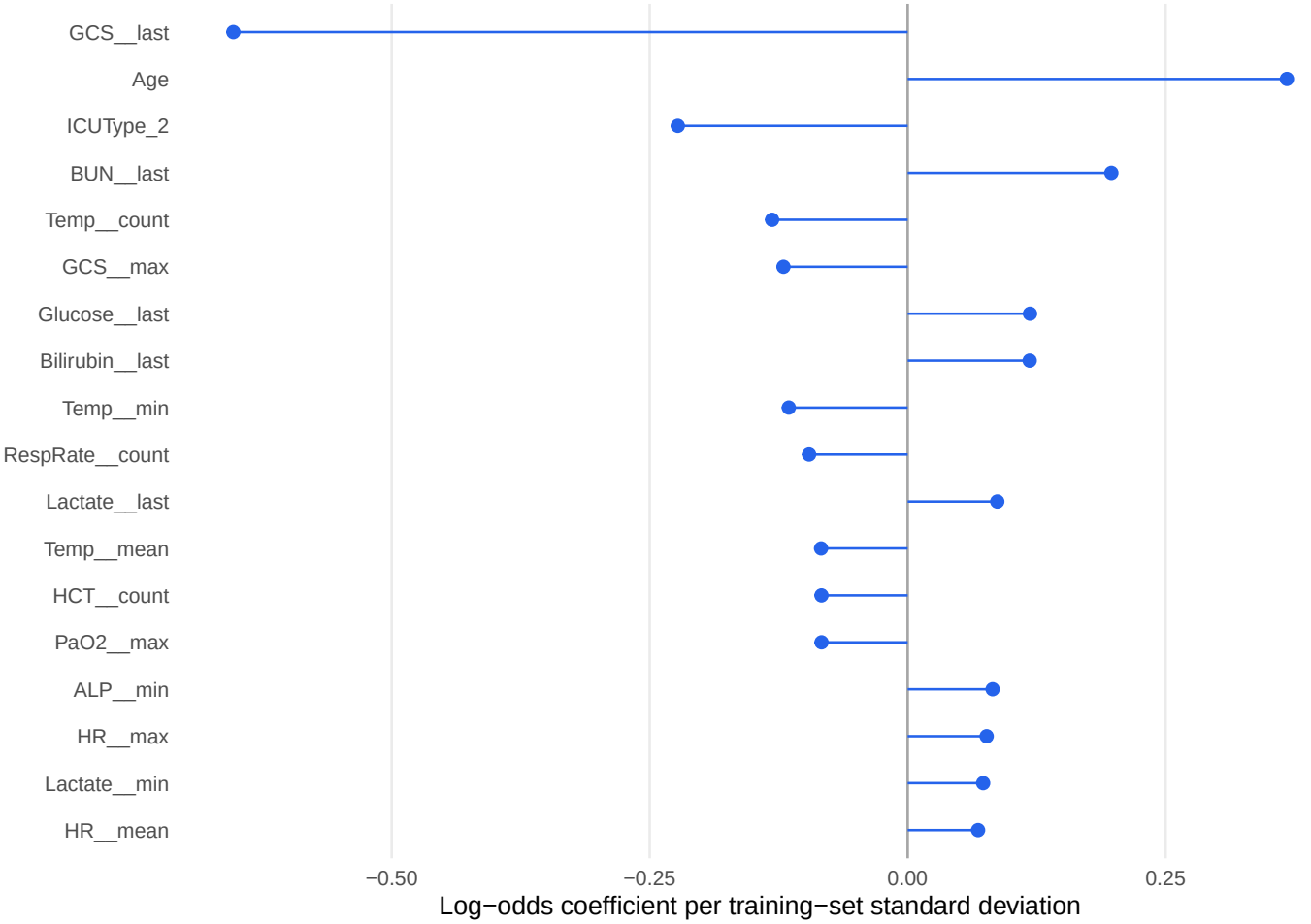


Table 5. Selection stability for the largest coefficients

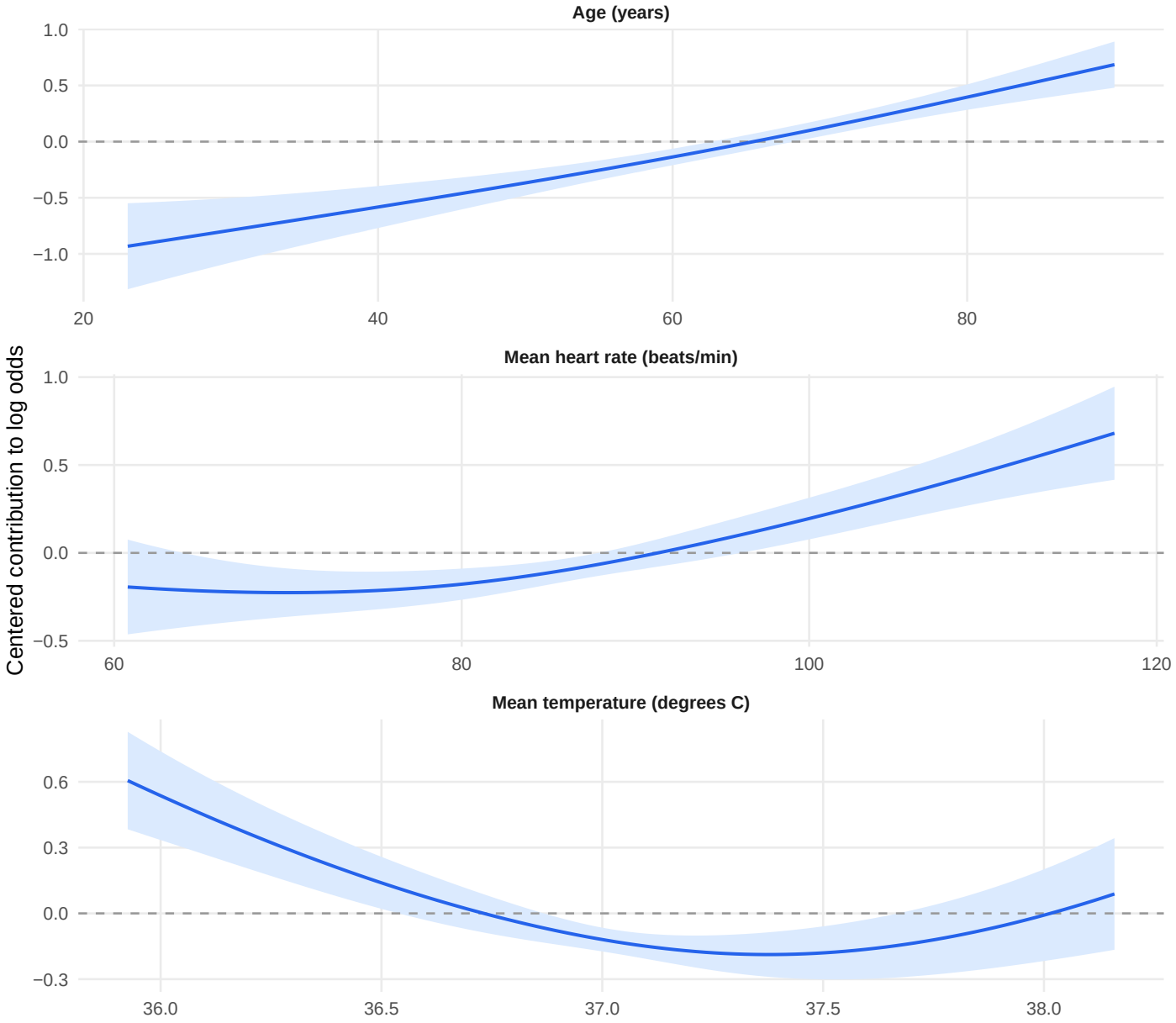
Predictor	Coefficient	Folds selected / 5
GCS__last	−0.653	5
Age	0.368	5
ICUType_2	−0.223	5
BUN__last	0.197	5
Temp__count	−0.131	5
GCS__max	−0.120	5

34 of the 55 full−data selected inputs appear in all five outer−fold fits. Correlated summaries can replace each other, so unstable selection limits interpretation.

2.5 Nonlinear effects

The 12 smooth inputs were chosen to cover age and major organ function: Age; mean HR, MAP, and Temp; minimum GCS, pH, and PaO2; maximum Creatinine, BUN, Lactate, and Glucose; and mean Urine. Figure 3 shows three centered contributions to log odds, with approximate 95% confidence bands. The x ranges cover the 2.5th to 97.5th observed percentiles. These are adjusted associations; zero is a centered reference, not an absolute risk of zero.

Figure 3. GAM partial effects with approximate 95% bands



Age is mostly increasing (effective df 1.44). Heart rate is mostly increasing (effective df 2.17). Temperature is U−shaped (effective df 2.23). Higher age and fast heart rate are plausible severity markers. Higher risk at low temperature is also plausible in critical illness. Effective degrees of freedom above 1 suggest nonlinearity. These are associations, with uncertainty at extreme values.

Final model and limitations

Choice and refit

I chose Elastic net, with mean outer−fold AUROC 0.843 (SD 0.005). My rule first finds the largest mean AUROC, then accepts methods within one standard error of that mean (cutoff 0.840). Among eligible methods, it favors elastic net, then GAM, then random forest for simplicity. This is a model−selection heuristic, not a statistical equivalence test.

After choosing the family, I repeated the three−fold parameter search on all set−a and refit the model on all 4,000 stays. The saved cp2_final_model.rds contains the model, feature order, imputation medians, scaling values, and retained columns. Checkpoint 3 can load this frozen pipeline. cp2_oof_predictions.csv saves held−out predictions for all five methods.

What the results do and do not establish

Inner tuning avoids choosing parameters with the outer validation outcomes. However, the same outer results choose the final model family, so its reported winning performance may still be optimistic. Folds are from one data source and do not test transport to a new hospital. Median imputation is simple but may lose detail. Some remaining extreme values may be measurement errors. Counts and ventilation can reflect care decisions as well as illness. A predictor being important does not mean that intervening on it would reduce mortality. Calibration and clinical usefulness will need separate evaluation in Checkpoint 3.

Validation and reproducibility

Automated checks cover unique stays, disjoint training and validation IDs, one held−out prediction per method and stay, probability bounds, and identical predictions after saving and reloading the final pipeline. The checkpoint2 pixi task rebuilds all derived files and this PDF. Model and table outputs remain local; the PDF is committed with the source code.

References

Silva I et al. (2012). Predicting in−hospital mortality of ICU patients. Computing in Cardiology 39:245−248. Data: https://physionet.org/content/challenge−2012/1.0.0/. Course checklist: https://steverozen.github.io/biostat707_2026/checkpoint2−3−checklist.html.

Appendix: retained predictors (1/2)

Table A1 lists every nonzero elastic−net coefficient, ordered by absolute size. The fold count shows how often each input was selected across the five outer fits. Predictors not listed have coefficient zero in the refit model.

Predictor	Coefficient	Folds selected / 5
GCS__last	−0.6534	5
Age	0.3675	5
ICUType_2	−0.2227	5
BUN__last	0.1974	5
Temp__count	−0.1314	5
GCS__max	−0.1203	5
Glucose__last	0.1186	5
Bilirubin__last	0.1183	5
Temp__min	−0.1152	5
RespRate__count	−0.0955	5
Lactate__last	0.0869	5
Temp__mean	−0.0839	5
HCT__count	−0.0835	5
PaO2__max	−0.0835	5
ALP__min	0.0824	5
HR__max	0.0766	5
Lactate__min	0.0732	4
HR__mean	0.0682	5
WBC__last	0.0638	5
Temp__last	−0.0633	4
Bilirubin__min	0.0626	5
Weight	−0.0621	5
pH__mean	0.0598	5
BUN__min	0.0584	5
Albumin__max	−0.0562	5
NIDiasABP__mean	−0.0498	4
NIMAP__last	−0.0460	4
FiO2__last	0.0447	5

Appendix: retained predictors (2/2)

Table A1 lists every nonzero elastic−net coefficient, ordered by absolute size. The fold count shows how often each input was selected across the five outer fits. Predictors not listed have coefficient zero in the refit model.

Predictor	Coefficient	Folds selected / 5
Urine__max	−0.0440	5
FiO2__min	0.0423	5
HR__last	0.0420	5
TroponinI__min	0.0390	4
pH__last	−0.0368	5
GCS__min	0.0357	5
Weight__max	−0.0314	3
GCS__count	0.0312	5
Na__count	0.0311	3
Urine__last	−0.0294	4
AST__min	0.0291	4
DiasABP__last	−0.0288	5
NIDiasABP__last	−0.0287	5
TroponinI__count	0.0267	5
TroponinT__count	0.0241	5
Albumin__mean	−0.0194	3
HCT__max	−0.0142	4
TroponinI__last	0.0125	4
ALT__count	0.0123	2
RespRate__mean	0.0117	4
NISysABP__last	−0.0106	4
HCO3__last	−0.0084	2
Lactate__mean	0.0075	3
PaCO2__max	−0.0037	3
Glucose__max	0.0034	2
Platelets__max	−0.0011	3
Glucose__count	0.0002	1